



The menace of colistin resistance across globe: Obstacles and opportunities in curbing its spread

P.S. Seethalakshmi^a, Riya Rajeev^a, Anushara Prabhakaran^a, George Seghal Kiran^b, Joseph Selvin^{a,*}

^a Department of Microbiology, Pondicherry University, Puducherry 605014, India

^b Department of Food Science and Technology, Pondicherry University, Puducherry 605014, India

ARTICLE INFO

Keywords:

Colistin resistance
Mcr
Residual colistin
Pan drug resistance
One health

ABSTRACT

Colistin-resistance in bacteria is a big concern for public health, since it is a last resort antibiotic to treat infectious diseases of multidrug resistant and carbapenem resistant Gram-negative pathogens in clinical settings. The emergence of colistin resistance in aquaculture and poultry settings has escalated the risks associated with colistin resistance in environment as well. The staggering number of reports pertaining to the rise of colistin resistance in bacteria from clinical and non-clinical settings is disconcerting. The co-existence of colistin resistant genes with other antibiotic resistant genes introduces new challenges in combatting antimicrobial resistance. Some countries have banned the manufacture, sale and distribution of colistin and its formulations for food producing animals. However, to tackle the issue of antimicrobial resistance, a one health approach initiative, inclusive of human, animal, and environmental health needs to be developed. Herein, we review the recent reports in colistin resistance in bacteria of clinical and non-clinical settings, deliberating on the new findings obtained regarding the development of colistin resistance. This review also discusses the initiatives implemented globally in mitigating colistin resistance, their strength and weakness.

1. Introduction

Polymyxins are a class of cyclic oligopeptide nonribosomal antimicrobials classified into five chemically distinct compounds i.e., polymyxins A, B, C, D, and E. Of these, colistin (polymyxin E) and polymyxin B have commercial market value (El-Sayed Ahmed et al., 2020; Dijkmans et al., 2015; Falagas et al., 2005; Son et al., 2019). Colistin is a cyclic deca-peptide antibiotic produced by the soil bacterium *Paenibacillus polymyxa* subsp. *Colistinus* (Koyama, 1950; Dijkmans et al., 2015). Colistin is a narrow spectrum antibiotic with bactericidal activity towards the members of Enterobacterales, but are inactive towards Gram-positive cocci, Gram-positive bacilli and Gram-negative cocci (Tan and Ng, 2006). Colistin mainly acts similar to that of a surfactant on the negatively charged lipopolysaccharide (LPS) of bacteria (Bialvaei and Samadi Kafil, 2015), where the positively charged colistin binds to the lipid A part of LPS layer and displaces cations like Mg^{2+} and Ca^{2+} , disrupting the integrity of outer membranes (Martis et al., 2014). Insertion of D-Leu⁶-L-Leu⁷ fragment or hydrophobic terminal fatty acyl

chain of colistin into the outer membrane results in the expansion of the outer membrane monolayer (Rhouma et al., 2016). The antibiotic disrupts the outer membrane and passes itself across by means of a self-promoted uptake mechanism (Hancock and Scott, 2000; Straus and Hancock, 2006). Colistin acts on the inner membrane by extending across the interface of fatty acyl chains and hydrophilic head groups, leading to its thinning, which eventually sets in cell lysis and cell death (Velkov et al., 2010). Although, in multi-drug-resistant *Acinetobacter baumannii* from clinical sources, colistin shows bactericidal activity due to the production of hydroxyl radicals (Sampson et al., 2012).

Colistin was used, approved by US FDA in 1959, for therapeutic purposes in clinical settings (El-Sayed Ahmed et al., 2020). However, it was dissuaded due to its neurotoxic and nephrotoxic side-effects (Bosso et al., 1991). Although colistin resistant bacterial infections emerged in clinical and non clinical setting, due to the emergence of extensively drug resistant Gram-negative bacteria and the scarcity of novel antibiotics in drug discovery pipeline, colistin was reinstated for clinical purposes (Li et al., 2006; Nang et al., 2019; Srinivas and Rivard, 2017;

* Corresponding author.

E-mail addresses: vimisha.ps@gmail.com (P.S. Seethalakshmi), rheanambiar9@gmail.com (R. Rajeev), anushara0010@gmail.com (A. Prabhakaran), segchalkiran@gmail.com (G.S. Kiran), josephselvinss@gmail.com, jselvin.mib@pondiuni.edu.in (J. Selvin).

<https://doi.org/10.1016/j.micres.2023.127316>

Received 1 May 2022; Received in revised form 27 November 2022; Accepted 1 February 2023

Available online 10 February 2023

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Centers for Disease Control and Prevention CDC, 2013; Ortwine et al., 2015). However, it has been under therapeutic and prophylactic use in animal husbandry for decades (Kempf et al., 2016). As per reports, by 2014, 96% of global colistin sulfate usage is attributed to chicken and pig husbandry industries (QYResearch Medical Research Centre, 2019). The indiscriminate use of colistin in livestock farming and aquaculture settings has resulted in a hotspot of colistin resistant bacteria which are pathogenic to humans (Liu et al., 2016; Shen et al., 2018). Therefore, a complete understanding of colistin resistance mechanisms is very crucial to develop new potential antimicrobials against notorious clinical Gram-negative bacteria (Olaitan et al., 2014b). In this review, we discuss the mechanisms of colistin resistance identified recently. In addition, the review analyses the success and limitations of the currently adopted action plan in combating antimicrobial resistance (AMR).

2. Resistant mechanisms involved in colistin resistance in bacteria

Colistin resistance in Gram-negative bacteria is complicated as they deploy a diverse set of resistance mechanisms to defend themselves from colistin (Aghapour et al., 2019). Some bacteria such as *Hafnia* sp., *Proteus mirabilis*, *Burkholderia* sp. and *Serratia marcescens* are resistant to colistin inherently (Olaitan et al., 2014b; Jayol et al., 2017), while others such as *A. baumannii*, *Pseudomonas aeruginosa*, and *Klebsiella pneumoniae* acquire resistance towards colistin (Olaitan et al., 2014b). Colistin resistance in Gram-negative bacteria arises due to changes in the outer LPS layer of the outer membrane as a result of post-translational modification (Jeannot et al., 2017). The modification of LPS is regulated by a variety of two-component systems in the bacterial chromosome (Jeannot et al., 2017). The changes involved in the regulation of such genes (*pmrAB*, *phoPQ*, and *mgrB*) was believed to be the only cause of colistin resistance in bacteria (Olaitan et al., 2014a) up until colistin resistance due to mobilization of *mcr* genes via plasmid was identified for the first time in 2016 (Liu et al., 2016). Therefore, colistin resistance in bacteria is either encoded by chromosomal mutations or by plasmids.

2.1. Colistin resistance encoded by the bacterial chromosome

Occurrence of huge variety of polymyxin resistance mechanisms indicates that the polymyxin resistance has been evolving over time. The negatively charged lipid A component of LPS on the Gram-negative bacteria is the initial binding site for colistin. Neutralization of the negative charge on outer membrane, production of excess capsular polysaccharide, activation of efflux pump systems and complete loss of LPS production are the main colistin resistance mechanisms encoded by the bacterial chromosome (Moffatt et al., 2019).

The two-component system *PmrA/PmrB* and *PhoP/PhoQ* is activated at sub-inhibitory concentrations of colistin (Ly et al., 2012). The *PhoP/PhoQ* two-component system activates *PmrA/PmrB* system via a connector protein *PmrD* (Poirel et al., 2017). The *PmrA* protein gets activated upon phosphorylation carried out by *PmrB* and this results in the transcription of *pmrC* that encodes phosphoethanolamine phosphotransferase (*PmrC*). This enzyme adds phosphoethanolamine (pEtN) moiety to the lipid A part of LPS (Gunn, 2008) which causes a drop in net charge of the outer membrane from -1.5 to -1 (Nikaido, 2003; Olaitan et al., 2014b). Over expression of *PmrA/PmrB* and *PhoP/PhoQ* can also lead to an enhanced expression of *pmrH* which results in the synthesis of N4-aminoarabinose (L-Ara4N), a positively charged sugar moiety (Lee and Ko, 2014). Binding of L-Ara4N to lipid A part of LPS in bacteria neutralizes the negative charge of the outer membrane to zero, thereby hindering the binding of colistin to the outer membrane (Ly et al., 2012; Nation, Li, 2009; Biswas et al., 2012; Trimble et al., 2016; Nikaido, 2003; Olaitan et al., 2014b).

The presence of colistin is shown to exalt the expression level of *pmrB* and *pmrC* by 8 and 7 folds in *A. baumannii* clinical isolates (Ghahraman et al., 2020). Even single amino acid substitutions in *PmrA/PmrB* system

can contribute to high level of colistin resistance in bacteria. In some clinical strains of *K. pneumoniae* with G250C substitution in *pmrB*, the minimum inhibitory concentration (MIC) of colistin was found to be higher than 128 $\mu\text{g/mL}$ (Azam et al., 2021). The activation of the *pmrF* operon, *ugd*, and *pmrC* which encodes Ara4N biosynthetic enzymes, UDP-glucose dehydrogenase and lipid A phosphoethanolamine transferase, respectively, leads to the production of L-Ara4N or pEtN which modifies lipid A in *Salmonella enterica* and *P. aeruginosa* in a *PmrA*-regulated resistance mechanism (Adams et al., 2009; Lee et al., 2004; Gunn et al., 1998; Ernst et al., 1999).

MgrB is a broadly conserved regulatory transmembrane peptide encoded by *mgrB*, that takes part in the negative feedback control of *PhoQ* circuit by inhibiting phosphorylation of *PhoP* either by the induction of phosphatase activity or suppressing the *PhoQ* kinase activity (Lippa and Goulian, 2009). Mutation in *PmrA* gene can also attribute to virulence of the bacteria by enabling them to survive in human neutrophils (Roland et al., 1993). Insertional inactivation and deletion of *mgrB* results in colistin resistance due to the upregulation of *PhoQ/PhoP* system (Cannatelli et al., 2013). *ParR-ParS* is a two-component regulatory system responsible for cationic peptides induced adaptive resistance in *P. aeruginosa*. Lipopolysaccharide modification operon (*arnBCADTEF*) is activated by *ParR-ParS* in the presence of cationic peptides, which leads to enhanced resistance against polymyxins. The adaptive resistance displayed by *P. aeruginosa* has serious clinical implications and indicates the possibility of using cationic antibiotics in great initial and longer interval bolus dosing as a treatment strategy (Skiada et al., 2011).

In some clinical isolates of *K. pneumoniae*, *CrrAB*, a two-component system, confers resistance to colistin by regulating the gene encoding a putative glycosyltransferase (H239_3059) associated with LPS modification (Wright et al., 2015). *CrrB* mutations induce over expression of *PmrAB*, *PhoPQ* and *KexD* multidrug efflux pump resulting in colistin resistance (Sun et al., 2020b), indicating that antibiotic resistance in bacteria could be a product of activation of multiple interconnected gene regulating systems in bacteria. A recent finding by Janssen et al. (2020) concludes that mutation in the two-component signal transduction system *basRS* (*PmrAB* in other Enterobacteriaceae) can also confer colistin resistance in *Escherichia coli*. Chromosomes harboring mobile genetic elements can aid the transmittance of resistant genes amongst different species and can result in the failure of colistin treatment. Presence of insertion elements in *mgrB* and *CrrB* in *K. pneumoniae* was found to rise colistin resistance frequencies significantly (Yang et al., 2020) and transform sensitive phenotypes to resistant (Zhang et al., 2021) (Fig. 1).

Mutational changes interfering with LPS biosynthesis can also prompt colistin resistance. The complete loss of LPS, along with lipid A, is a very fascinating polymyxin resistance mechanism identified in *A. baumannii* (Moffatt et al., 2019). Loss of lipid A resulting from mutational changes in the genes *LpxA*, *LpxC*, and *LpxD* involved in lipid A biosynthesis, can also prevent the binding of colistin to LPS (Moffatt et al., 2010). This resistance mechanism is at present observed only in *A. baumannii* (Moffatt et al., 2019). Any mutations in *lptA*, a *pmrC* ortholog responsible for lipid A phosphoethanolamine decreases colistin resistance in *Neisseria meningitidis*. This indicates that this is the only major colistin mechanism in *N. meningitidis* (Tzeng et al., 2005; Adams et al., 2009). Recently Sun et al. (2020a) reported that in certain *A. baumannii* strains, mutations in the post-transcriptional regulator *miaA* synergistically enhanced colistin resistance mediated by *PmrA* mutation.

LPS biosynthesis is also impaired due to mutational changes in the *galETK* repressor KP1_0061, conferring resistance to colistin (Choi and Ko, 2020). Although colistin resistance mediated by chromosomal mutations is most commonly reported in Gram-negative bacteria, gene mutations giving rise to colistin resistance in *Vibrio* sp. have also been detected. Mutation in the metabolic gene *aceF*, that encodes pyruvate dehydrogenase reduced pyruvate cycle and the expression of

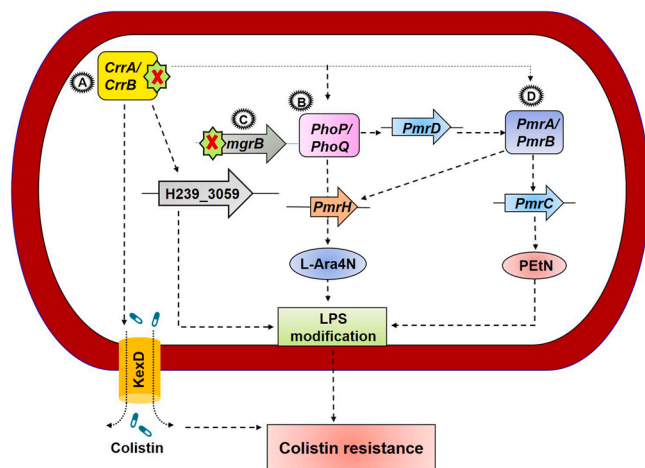


Fig. 1. : Colistin resistance mediated by chromosomal mutations in two component systems. **A.** *CrrAB* induces LPS modification via glycosyltransferase H239_3059 and mutational changes in *CrrAB* enhances the efflux of colistin by overexpression of KexD multidrug efflux pump. **B.** *PhoP/PhoQ* two-component system activates *PmrA/PmrB* resulting in the synthesis and addition of phosphoethanolamine to LPS. **C.** Inactivation or deletion of the *mgrB*, elevates the expression of *PhoP/PhoQ* which results in LPS modification **D.** *PhoP/PhoQ* and *PmrA/PmrB* upregulates *pmrH* leading to the synthesis and incorporation of N4-aminoarabinose to LPS.

sodium-translocating NADH-ubiquinone oxidoreductase in *Vibrio alginolyticus*, which subsequently changed the membrane potential and reduced the binding of colistin to LPS (Li, L et al., 2020). The modification of LPS can influence the sensitivity of bacteria to other unrelated antibiotics (Saleh et al., 2020). This can specifically hamper the synergistic treatment combinations of colistin with other antibiotics.

One of bacterial outer membrane modification, enhanced production of capsular polysaccharides have been reported to mediate polymyxin resistance in bacteria like *K. pneumoniae*. Capsular polysaccharides reduces the possibility of colistin interaction with target sites (Campos et al., 2004; Falagas et al., 2010). The capsule of *K. pneumoniae* was found to be multilayered in colistin resistant strains than in sensitive cells (Formosa et al., 2015). Purified capsular polysaccharide when supplemented to un-capsulated *K. pneumoniae* strain showed increase in polymyxin resistance (Llobet et al., 2008; Moffatt et al., 2019).

Heteroresistance is another complex colistin resistance mechanism found in *Enterobacter* spp. where only a sub population of a strain becomes resistant, leading to the failure of antibiotic therapy in patients (Band et al., 2016). Heteroresistance to colistin in *E. cloacae* complex is mediated by a membrane protein encoded by *ecr* that upregulates *phoP* and the *arnBCADTEF* operon (Huang et al., 2019). Global transcription system *soxR/soxS*, was also proven to be involved in colistin resistance mediated by efflux pump proteins *acrAB-tolC* in *E. cloacae* (Telke et al., 2017). The exact reason behind elevated levels of *soxRS* in heteroresistant *E. cloacae* is a puzzle yet to be resolved. The absence of changes in the expression levels of *mgrB*, *pmrC* and *arnA* genes in heteroresistant isolates hints at the possible existence of a different colistin resistance mechanism in *Enterobacter* sp. (Telke et al., 2017).

Besides mutational changes, cryptic phages in the chromosome can also confer antibiotic resistance in bacteria (Wang et al., 2010). *K. pneumoniae* experimentally grown for 30 days in the presence of colistin showed amplification of a sequence homologous to a prophage (Enterobacteria phage Fels-2) of size 34 Kb suggesting the possible involvement of retrons and phages in the development of colistin resistance (Sánchez et al., 2021). Further investigation is required to understand the molecular basis of such phage-mediated colistin resistance mechanisms in bacteria (Fig. 2).

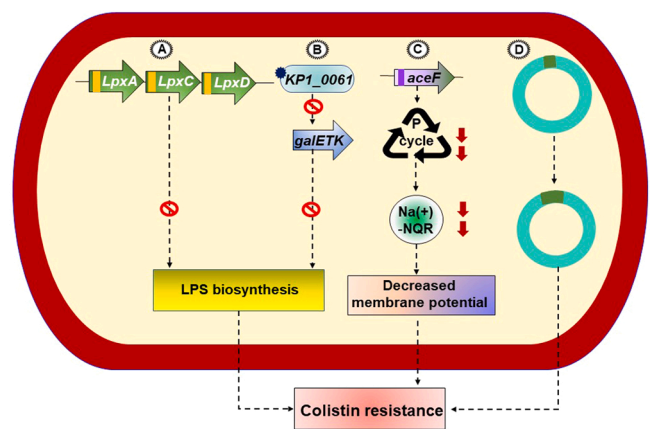


Fig. 2. : Colistin resistance mediated by chromosomal mutations in genes other than two component systems. **A.** Mutations in *LpxA*, *LpxC*, and *LpxD* prevents the formation of LPS. **B.** Mutational changes in the repressor protein KP1_0061 repress the *galETK* operon obstructing LPS biosynthesis. **C.** The membrane potential is altered due to reduced expression of sodium-translocating NADH-ubiquinone oxidoreductase resulting from mutational changes in *aceF*. **D.** Amplification of *Enterobacteria* phage Fels-2 in chromosome conferring colistin resistance.

2.2. mcr and transmittable colistin resistance

The transmission of colistin resistance via plasmids in animals and humans was first documented in China, encoded by the gene *mcr-1* (Liu et al., 2016). Since then, genetic variants of *mcr-1* have been reported across the globe (Di Pilato et al., 2016; Yang et al., 2017), indicating the presence of an ongoing evolution of the gene (Sun et al., 2018). Currently, there are ten different variants of *mcr* (*mcr-1* – *mcr-10*) identified from bacteria originating from humans, animals, foods, natural environments and farms (Hussein et al., 2021). The fast dissemination of *mcr-1* can be attributed to the indiscriminate use of colistin for veterinary purposes (Shen et al., 2016). Even though most studies have reported the presence of *mcr* gene in plasmids, their presence in the chromosome has also been described (Zurfluh et al., 2016; Xu et al., 2021). *mcr-1* encodes for the enzyme MCR-1 (phosphatidyl-ethanolamine transferase) which lead to the incorporation of pEtN to lipid A part of the LPS structure and bacteria harboring this gene can exhibit high level of resistance to colistin (Poirel et al., 2017). Recent findings suggest that *mcr* gene elicits changes in the physical attributes of the membrane, such as rigidity, fluidity and tension (Ma et al., 2021). MCR-4 exhibits similar enzymatic activity to that of MCR-1; however, *mcr-4* and *mcr-3* gene can also impede the ROS accumulation triggered by colistin (Xu et al., 2018; Zhang et al., 2019). MCR-1 can compromise the membrane integrity, thereby reducing the sensitivity of bacteria to hydrophobic antibiotics such as rifampicin, gentamicin and kanamycin (Li, B et al., 2020). MCR-1 enables the addition of pEtN to LPS with efflux pump proteins (Li et al., 2019). Li et al. (2021) identified that the multidrug efflux pump membrane proteins AcrA and TolC as the key interacting proteins of MCR-1, confirming the involvement of the two-component system AcrA/TolC in mediating colistin resistance. The AcrAB–TolC efflux pump bestows antimicrobial resistance to a wide range of antibiotics by permitting the transport of chemical compounds with little chemical similarities out of the cell (Du et al., 2014). Padilla et al. (2010) found that AcrAB–TolC efflux pump of *K. pneumoniae* cells exhibited resistance to polymyxin in an energy-dependent manner (Fig. 3).

The role of *mcr* in the global dissemination of colistin resistance is supported by a vast number of studies. Yet, in some cases a correlation between *mcr* and colistin resistance was not ascertained. *Salmonella enterica* and *E. coli* isolates harboring *mcr-9* were found susceptible to colistin at a MIC less than 1 µg/mL (Tyson et al., 2020). It is also

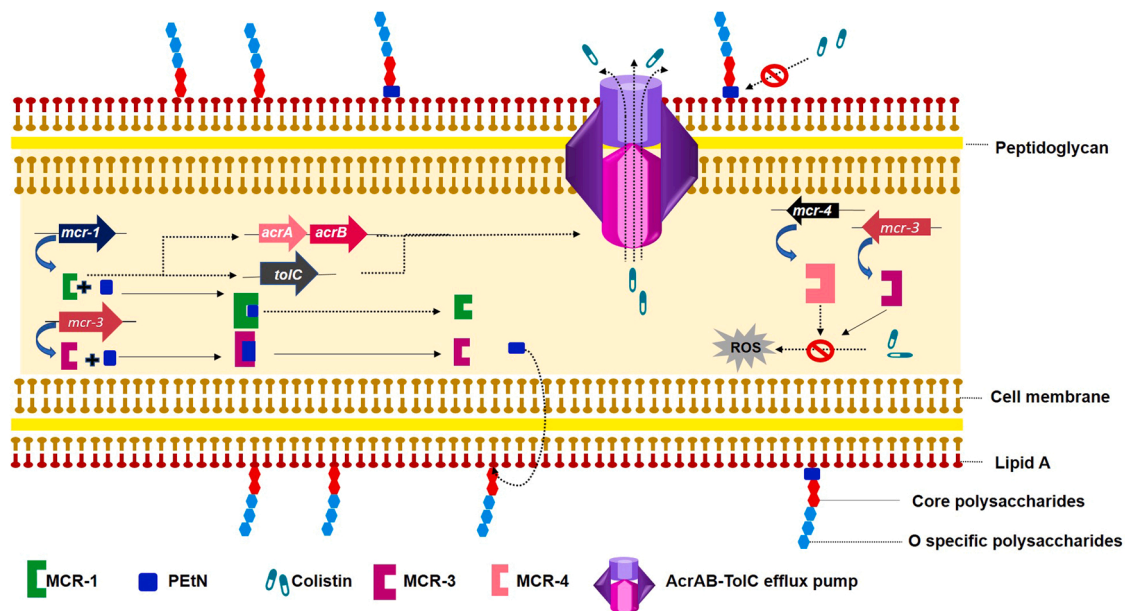


Fig. 3. : Mechanisms involved in colistin resistance mediated by *mcr*. **A.** *mcr-1* encodes the enzyme MCR-1 which catalyses the addition of phosphoethanolamine to lipid A portion of LPS **B.** MCR-1 interacts with AcrAB-TolC efflux pump resulting in the efflux of colistin **C.** MCR-4 can inhibit the accumulation of ROS mediated by colistin.

speculated that MCR is ubiquitously present in the bacteria of natural environment as a defense system against bacteriophages (Khedher et al., 2020). This complies with the fact that *Serratia* sp. and *Proteus* sp. possesses *mcr* variants and are inherently resistant to colistin (Khedher et al., 2020).

3. Colistin resistant bacteria from clinical and non-clinical entities

Colistin resistance genes have been broadly characterized in humans, diverse barnyards, and the poultry sector globally (Apostolakis, Piccirillo, 2018; Liu et al., 2016). Recent reports suggest their abundance in aquaculture settings and sewage treatment plants (Chen et al., 2017; Wang et al., 2019). Colistin resistant bacteria bearing *mcr* genes have been detected in meat products, wildlife, agricultural soil, waste water, and even natural water bodies (Liakopoulos et al., 2016; Sabala et al., 2021; Lopes et al., 2021; Hayashi et al., 2021; Teng et al., 2021) implying the rapid dissemination of colistin resistant genes to different environments.

The gene pool of colistin resistant or other resistant genes in the environment is mainly due to two scenarios. Either they are hosted by normal microbial flora in low antibiotic concentration or accomplished by bacteria during high antibiotic selection pressure (Lv et al., 2021). The live poultry markets, their environment and the workers act as important reservoirs for a plethora of antibiotic resistant genes, including *mcr* and continuous exposure of workers to poultry environment is found to change the distribution of their gut microbiota and resistome (Wang et al., 2021). Wang et al. (2021) also demonstrated that the live poultry systems are even more efficient in developing, spreading *mcr* and other genes from farms to the environment as well as from non-pathogenic bacteria to opportunistic pathogenic bacteria. Amin et al. (2020) reported *mcr-1* positive isolates of *E. coli* possessing double replicon type plasmid along with supplementary *bla*_{CTX-M} group-1, *bla*_{TEM-1}, *qnrB*, *rmtB* genes from poultry sources. The estimated number of *mcr* gene carrying bacterial strains is higher in farm settings than the hospitals (Liu et al., 2016). The presence of *mcr* genes in animal settings, poultry and poultry products is identified as the reason for the invasion of *mcr* genes carrying Enterobacterales into humans. AMR is an emerging threat to the food system (Selvin et al., 2020) and the presence

of *mcr-1*-positive isolates in raw food samples (Ghafur et al., 2019) infers the chances of potential food-borne outbreaks of AMR pathogens.

Nguyen et al. (2020) suggest that untreated domestic wastewater bearing human fecal contaminants is the main driving factor contributing to the *mcr-1* gene dissemination in urban aquatic environments of Vietnam. Along with *mcr-1*, other clinically significant Antibiotic Resistance Genes (ARGs) such as *ermB* gene (erythromycin resistance), *tetM* (tetracycline resistance) and β -lactam resistance genes (*CTX-M-32*, *bla*_{TEM}, *CMY-2*, and *CTX-M*) were detected from wastewater effluent of treatment plants in Germany (Hembach et al., 2017). *mcr* gene variants such as *mcr-3*, *mcr-4*, *mcr-5*, *mcr-7* were predominant in German municipal waste waters compared to *mcr-1* (Kneis et al., 2019). The major proposed causes of the appearance of certain ARGs in wastewater environments include a hike in nutrient accessibility, increased cell density in activated sludge, heavy metal co-selection and selection pressure exerted by xenobiotics (Rizzo et al., 2013; Berendonk et al., 2015). Even after treatments like the activated sludge process, there is no complete removal of ARGs from water and thus, a heavy load of *mcr* genes arrives in aquatic ecosystems (Pärnänen et al., 2019). *mcr* in water environments is worrisome as it can enter the food chains of humans and animals since polluted water is used in agriculture and livestock industries (Nguyen et al., 2020). Lopes et al. (2021) reported the presence of zoonosis-associated *E. coli* ST131-H22 with its resistome, encoding *mcr-1.1*, *bla*_{CTX-M-15}, and *qnrB19* in the soil. The agro-ecosystems can transmit ARGs to vegetables exposing humans to clinically important pathogens (Reid et al., 2020). The spread of *mcr-1* in vegetables documented in China was found to be via IncX4, or IncI2 plasmids and these plasmids are similar to plasmids from animals and clinical samples from different countries (Liu et al., 2019). Vegetables with *mcr* genes may not have any direct consequences, but the gut microbiome may receive ARGs through horizontal gene transfer (Maeusli et al., 2020).

Migratory birds can also act as a super-spreaders in the transmittance of *mcr* across different environments ranging from landfills to rice fields or water supplies, posing a threat to humans and wildlife (Jarma et al., 2021). Studies have pointed out that long-range migratory harboring *mcr* could contribute to the dispersal of colistin resistant bacteria across continents (Liakopoulos et al., 2016; Mohsin et al., 2016). Colistin resistance reported in wildlife species could result from the acquisition of resistant genes from other ecosystems rather than selective pressure,

as wildlife is not exposed to colistin. In addition to being an environmental reservoir of resistant bacteria, the presence of ARGs in wildlife can have grave repercussions. This includes the zoonotic transmission of enteric pathogens to humans and issues related to the medical treatment of wildlife (Radhouani et al., 2014) (Table 1).

Many researchers have reported nosocomial infections in patients caused by colistin resistant organisms (Yoon et al., 2022; Snyman et al., 2021a). Selection pressure by the colossal administration of colistin and clonal expansion created a number of cases of multidrug resistant bacterial infections (El-Bogdanovich et al., 2011; El-Mokhtar et al., 2021). However, this finding cannot be generalized as studies have reported colistin resistant bacterial infections without a record of colistin administration (Olaitan et al., 2016a, 2016b). One of the serotypes of *Salmonella enteritidis* from blood stream displayed more resistance to colistin compared to *Salmonella typhimurium* isolated from the intestine. The colistin susceptibility, along with the mechanism of resistance, was found to be dependent on serotypes (Luo et al., 2020). This stresses the need for sero-specific cut-off values to determine the MIC in *Salmonella* sp.

4. Residual colistin: a hassle in fighting colistin resistance

The use of colistin as a veterinary pharmaceutical regimen for the treatment of infectious diseases can have many profound implications apart from antimicrobial resistance. The administration of colistin in food animals can contribute to the deposition of residues in milk and meat, which upon consumption, can result in neurotoxic and nephrotoxic effects in humans (Wang et al., 2019). The oral administration of colistin in food animals has been found to accumulate colistin residues at high levels in their gastrointestinal tracts and feces (Peng et al., 2021). The veterinary use of colistin can promote the development of colistin resistance in the gut microbiota of animals (Peng et al., 2021). Residual colistin is one of the chief factors that leads to persistence of *mcr-1* in the environment (Shi et al., 2021). The fecal matter of animals receiving colistin can contaminate the environment and aggravate the persistence of *mcr-1*. Polymyxin A and B have been detected even in cattle manures intended for agricultural purposes (Huygens et al., 2021), and hence,

raw manures could also act as a potential route for the transmission of colistin residues and resistant genes to agricultural fields. Colistin residues, while reaching agricultural fields from raw manures may accumulate in soil enabling its uptake by plants. However, not much attention has been given in this regard to draw conclusions. Still, the possibility that presence of colistin residues in plants may result in biomagnification and bioaccumulation cannot be ruled out (Tasho and Cho, 2016).

Intake of animal-based food products contaminated with trace amounts of antibiotics can result in the imbalance of human gut microbiota and upset intestinal health. The oral consumption of food harboring antibiotic residues disrupts the colonization barrier and metabolic functioning of intestinal microbiota (Cerniglia and Kotarski, 1999; Gorbach, 1993; Woodward, 1998). Jeong et al. (2009) found that most bacteria originating from the fecal sample were resistant to colistin sulphate except *E. coli*. However, a comparison of *E. coli* of ATCC-type culture and *E. coli* of intestinal origin showed significant differences in terms of susceptibility to colistin sulphate. *E. coli* from fecal samples were two times less sensitive than ATCC type culture, suggesting the possible development of resistance of the bacteria against colistin (Jeong et al., 2009). Exposure of colistin with other antibiotics can also have drastic implications in the human gut microbiota. A recent study by Li et al. (2021) concluded that the combinatorial exposure of colistin and amoxicillin altered the antibiotic resistome and enhanced the abundance of opportunistic pathogens in the human intestine. The Codex Alimentarius FAO-WHO has fixed the maximum residue limits (MRLs) of colistin at 50 and 150 µg/kg in milk and meat, respectively, which is safe for human consumption (Codex Alimentarius, 2017). However, the continued consumption of meat products contaminated with antibiotics can have serious health implications as they cause oxidative stress in the testes (Aksu et al., 2018) and can get accumulated in the brain and kidneys (Wang et al., 2016; Ma et al., 2009). Therefore, it is essential to completely restrict the use of colistin in food animals. Residual colistin emerging from effluents of antibiotic manufacturing industries and hospital waste can reach aquatic ecosystems. The role of such residues in developing resistomes in aquatic animals, humans and wildlife has not been documented yet and is worthy of investigation.

Table 1
Overview of colistin resistance reported from various environmental settings.

Organism	Gene conferring colistin resistance	Country	MIC of Colistin	Samples	References
<i>E. coli</i>	<i>mcr-1</i>	Lebanon	4–64 mg/L	Chicken samples	Al-Mir et al. (2021)
Chicken and human gut microbiome	<i>mcr-1</i>	China	-	Chicken and environmental samples from live poultry markets and broiler farms, human samples	Wang et al. (2021)
<i>E. coli</i>	<i>mcr-1</i>	Bangladesh	4–64 mg/L	Chickens and Farm Environmental samples	Das et al. (2021)
<i>E. coli</i>	<i>mcr-1.1</i>	Germany	4–8 mg/L	Poultry and pig slaughter house samples, Waste water	Savin et al. (2020)
<i>K. pneumoniae</i>			4– > 64 mg/L	treatment plants samples	
<i>K. pneumoniae</i>	<i>MgrB</i> Gene Alterations	Iran	>2 mg/L	Poultry diagnostic Centre and chicken slaughter house samples	Pishnian et al. (2019)
<i>E. coli</i>	<i>mcr-3</i>	Bangladesh	-	Poultry, House flies, Pond samples	Sobur et al. (2019)
<i>E. coli</i>	<i>mcr-1</i>	Tunisia	8 mg/L	Chicken samples	Maamar et al. (2018)
<i>E. coli</i>	<i>mcr-1</i>	Taiwan	4–8 mg/L	River samples	Teng et al. (2021)
<i>E. coli</i>	<i>mcr-1.1</i>	Lebanon	4–64 mg/L	Irrigational Water	Hmede et al. (2019)
<i>A. baumannii</i>	<i>mcr-1</i>	Iraq	128 µg/mL	Clinical specimens	Al-Hasnawy et al. (2021)
<i>Acinetobacter nosocomialis</i>	<i>mcr-4.3</i>	South Africa	16 mg/L	Clinical specimens	Snyman et al. (2021b)
<i>E. coli</i>	<i>mcr-1</i>	Czech Republic	4–8 mg/L	Clinical specimens	Tkadlek et al. (2021)
<i>E. coli</i>	<i>mcr-1</i>	Egypt	-	wild birds samples	Ahmed et al. (2019)
<i>Citrobacter braakii</i>	<i>mcr-1</i>	Bolivia	8 mg/L	Ready-to-eat food samples	Sennati et al. (2017)
<i>E. coli</i>	<i>mcr-1–5</i>	Belgium	> 2 mg/L	Pig, cattle and poultry feces	Timmermans et al. (2021)
Metagenomic approach	<i>mcr-3</i> <i>mcr-4</i> <i>mcr-5</i> <i>mcr-7</i>	Germany	-	Municipal waste water	Kneis et al. (2019)

5. Colistin resistance in pan drug resistant bacteria: an impending threat to public health

Magiorakos et al. (2012) defines Multi Drug Resistance (MDR) as acquired resistance of the organism to at least one agent in three or more antibiotics, Extensively Drug Resistant (XDR) as the resistance of the organism to at least one agent in all but two or fewer antibiotics and Pan Drug Resistance (PDR) as the resistance of the organism to all agents in all antibiotics. PDR infections signify a horrifying situation for health care as it leaves no choice of antibiotics for treatment. Several reports have mentioned colistin as the last line of antibiotic to be active against MDR pathogens (Levin et al., 1999; Garnacho-Montero et al., 2003; Michalopoulos et al., 2005; Falagas et al., 2006).

The most commonly reported variant of *mcr* gene, *mcr-1*, readily co-exists with *bla_{CTX-M}* gene and the selection pressure on *mcr-1* further selects for broad spectrum cephalosporin resistance (Yang et al., 2016). The co-existence of carbapenem resistance gene and colistin *mcr-1* in *Enterobacterales* is illustrated in many studies (Lai et al., 2017; Delgado-Blas et al., 2016). Do et al. (2020) reported the presence of multidrug resistant plasmid in pigs and consequent problems of ineffective treatment options for edema disease in weaned piglets. Timmermans et al. (2021) in their studies reported an *E. coli* isolated from cattle faeces with the triple occurrence of *mcr* gene (*mcr-1*, *mcr-3*, *mcr-5*) and carries 27 resistant genes along with 7 replicons. They also demonstrated that the *mcr* genes exist in various lineages of *E. coli*, chiefly in non-pathogenic isolates, in diverse plasmids of isolates hosting different AMR genes. These diverse AMR genes indicate that *mcr* genes are highly stable to co-exist with other AMR genes (Timmermans et al., 2021). Colistin resistant *K. pneumoniae* harboring *NDM-1* and *OXA-48* carbapenemases caused a fatal outbreak in a hospital at Turkey and were associated with high mortality rates in infected patients (Guducuoglu et al., 2018). Similar reports of colistin resistance in patients infected with XDR and PDR bacteria were reported in Egypt and Iran (Haeili et al., 2018; Abd El-Baky et al., 2020). In some extensively drug resistant *E. coli* clinical isolates which carried *mcr-1.1* located on IncI2 plasmid alongside other ARGs (*bla_{CTX-M-55}*, *bla_{NDM-1}*, *bla_{TEM}*, *qepA1*, *rmtB*), tige-cycline was found to be active against the pathogen (Han, S et al., 2020).

The conjugative properties of plasmid is responsible for the transmission of plasmid containing *mcr* genes between bacteria (Nang et al., 2019). Conjugative plasmids and insertion sequences effectively interact to improve the horizontal transfer of AMR genes in a complex environment (Che et al., 2021). According to a study, all the *mcr-1* elements being transmitted originate from the same initial mobilization of *mcr-1* by an IS*Apl1* transposon during the mid 2000 s. The demographic expansion which followed, is responsible for the present global distribution of *mcr* genes (Wang et al., 2018c). Mechanisms such as horizontal gene transfer, transposons, or jumping genes are mentioned as the reason for the co-existence of *mcr* gene with other ARGs (Wang et al., 2020a). The *mcr-1* positive *E. coli* isolates from the rivers in Taiwan exhibit the presence of Tn3 family transposon in the epidemic IncX4 plasmids (Teng et al., 2021). Peng et al. (2019) identified strains of *E. coli* harboring *mcr-1* and *bla_{NDM-1}* showing resistance to a wide range of antibiotics such as cephalosporins, aminoglycosides, β -lactam combination agents, fluoroquinolones, phenicol, tetracyclines, and sulfonamides from Pig Farms in China. Moreover, the study also concluded that the conjugative plasmid IncFII-type bearing *bla_{NDM-1}* *fosA3*, *bla_{TEM-1B}*, *rmtB* and IncX4-type with *mcr-1* of the isolates can co-conjugate transferring all the ARGs to recipient cells. Also, the presence of 16 S RMTase genes such as *rmtB* in Gram-negative bacteria may render novel antibiotics such as aminoglycoside plazomicin ineffective (Taylor et al., 2018). A study conducted by Huang et al. (2020) pointed out that the co-harboring strains were genetically diverse before colistin administration and eventually became genetically similar after the use of colistin. They also suggest that the increased amount of colistin in use can select for colistin-carbapenem resistant strains (Huang et al., 2020). The simultaneous occurrence of *Int-1* and *Int-2* in clinical isolates

co-expressing *mcr-1* with Extended Spectrum Beta Lactamase (ESBL) gene variants like *bla_{CTX-M-15}*, *bla_{CMY-2}*, *bla_{NDM-1}*, *bla_{NDM-5}* were reported by Ejaz et al. (2021). As mentioned in the review, detection of *mcr* in some cases may not confirm colistin resistance. A clinical isolate of *Enterobacter hormaechei* in which *mcr-9* co-occurred with *bla_{TEM-1}*, *bla_{VIM-4}*, *AmpC* β -lactamase, *fosA*, *qnrA1* was found to be susceptible to colistin (MIC, < 0.25 μ g/mL) and the susceptibility was associated with the absence of *qseC/qseB* regulatory system in the bacteria (Soliman et al., 2020). Although *mcr* is the commonly reported colistin resistant gene co-existing with other ARGs, colistin resistance in PDR and MDR could also emerge due to chromosomal mutations conferring resistance to colistin. This is justified in a study reported by Longo et al. (2019) where *K. pneumoniae* carrying *bla_{CTX-M}*, *armA* and *rmtB* were resistant to colistin due to mutations in *mgrB*, *pmrB*, *phoQ* and *crrB*.

Bacterial fitness plays a vital role in the persistence of bacteria in those environments which are devoid of antibiotics or carry trace quantities of antibiotics. Resistance encoded by chromosome or plasmid can impart fitness cost in bacteria (Andersson, 2006; Hershberg, 2017) by reduction of virulence, transmission (Borrell and Gagneux, 2009) or even growth of the bacteria (Song et al., 2014). *mcr-1* specifically hindered the growth of *K. pneumoniae* under competitive conditions while it had no effect in the fitness of *E. coli* carrying *mcr-1* which could be the reason why the gene has been frequently reported in the *E. coli* (Tietgen et al., 2018). In contrast to *mcr-1*, chromosome mediated colistin resistance demands a fitness cost in *E. coli* (Choi et al., 2020), highlighting the difficulty of curbing the spread of *mcr-1* bearing *E. coli* into other environments. Co-existence of *bla_{NDM-1}*, *bla_{NDM-4}*, *bla_{NDM-5}*, and *bla_{NDM-9}* along with *mcr-1* in *E. coli* had low fitness cost, suggesting the ease with which carbapenem resistant *E. coli* can evolve into PDR by acquiring colistin resistance (Wang R et al., 2018; Zhang et al., 2017) (Table 2).

6. Global initiatives to tackle colistin resistance in bacteria

AMR in bacteria is a burning issue with serious implications on human, animal and environmental health. Many studies have attested to the strong correlation between the use of antibiotics in food animals and the development of AMR in bacteria (Marshall, Levy, 2011; Aidara-Kane, 2012). Recent studies have detected the contamination of colistin residues and associated abundance of *mcr-1* genes in animal feedlots (Xia et al., 2019; Shi et al., 2021). In response to the emergence of plasmid borne *mcr-1* resistance, countries including China, India, Japan, Brazil, Argentina, Thailand and Malaysia have banned the use of colistin as growth promoting agent in food animals (Wang et al., 2020b). The European Medicines Agency brought the veterinary use of colistin to be permitted according to the recommendations of Antimicrobial Advice Ad Hoc Expert Group, limiting its use only in cases where no other alternative drugs are available (EMA, 2016). Following the ban of colistin in China, antibiotic production has decreased with significant reduction of colistin resistance in animals and humans (Wang et al., 2020b). Usui et al. (2021) also reported a significant decrease of *mcr-1* positive *E. coli* after the ban of colistin as feed additive for livestock in Japan. Despite the ban of colistin as feed additive in China, no restrictions have been enforced upon its therapeutic use in veterinary (Walsh and Wu, 2016). This may open new grounds for the indiscriminate use of colistin in poultry and aquaculture-based industries. The prohibition of colistin alone will not resolve the issues of colistin resistance as *mcr-1* dwells in hospital environments and communities, even in the absence of colistin (Tian et al., 2017; Wang et al., 2017).

The restrictions have been instigated only in some countries, while the use of colistin in poultry is still predominant in low and middle-income countries (LMIC) (Dandachi et al., 2018, 2020). This introduces a divide in the global efforts taken to mitigate colistin resistance. Import and export of food and foodstuffs may act as a route for dissemination of AMR to different countries. Therefore, a joint effort by all countries is required to effectively contain colistin resistance (Olaitan

Table 2

Overview of ARGs reported to coexist with genes associated with colistin resistance.

Resistant Gene	Coexisting ARGs	Host Organism	Source	Country	References
<i>mcr-1</i>	<i>bla_{NDM-5}</i>	<i>E. coli</i>	Clinical specimens	Japan	Uchida et al. (2018)
<i>mcr-1</i>	<i>bla_{CTX-M}</i>	<i>E. coli</i>	Chicken samples	China	Yang et al. (2017)
<i>mcr-1</i>	<i>bla_{CTX-M55}</i>	<i>Salmonella enterica</i>	Chicken samples	China	Yang et al. (2016)
<i>mcr-1</i>	<i>bla_{NDM-5}</i>	<i>E. coli</i>	Clinical specimens	U.S	Mediavilla et al. (2016)
<i>mcr-1</i>	<i>bla_{NDM}</i>	<i>E. coli</i>	Clinical specimens	China	Huang et al. (2020)
<i>mcr-9.1</i>	<i>bla_{IMP-4}</i>	Enterobacteriaceae	Clinical specimens	Australia	Maccesic et al. (2021)
<i>mcr-1</i>	<i>bla_{SHV}, bla_{TEM}, bla_{VIM}, bla_{NDM-1}, bla_{IMP}</i>	<i>K. pneumoniae</i>	Farm Environmental samples and clinical specimen	Egypt	Elmonir et al. (2021)
<i>mcr-1</i>	<i>bla_{KPC-2}</i>	<i>E. coli</i>	clinical specimen	China	Wang et al. (2020a)
<i>mcr-1</i>	<i>OXA-48</i>	<i>E. coli</i>	chicken samples and clinical specimen	Nepal	Muktan et al. (2020)
<i>mcr-1</i>	<i>bla_{NDM-1}</i>	<i>E. coli</i>	Pig and Farm Environmental samples	China	Peng et al. (2019)
<i>mcr-1.1</i>	<i>tet(X7)</i>	<i>E. coli</i>	Chicken samples	Egypt	Soliman et al. (2021)
<i>mcr-1</i>	<i>bla_{NDM}</i>	<i>E. coli</i>	livestock	China	Wang R et al. (2018)
<i>mcr-1</i>	<i>bla_{SHV}, bla_{TEM}</i>	<i>E. coli</i>	clinical specimen	Egypt	Zakaria et al. (2021)
<i>mcr-1</i>	<i>bla_{CTX-M-15}, bla_{TEM-1}, qnrB19</i>	<i>E. coli</i>	Barbary macaques	Algeria	Bachiri et al. (2018)
<i>mcr-1</i>	<i>rmtB, bla_{CTX-M}, bla_{CTX-M-9}, bla_{CTX-M-1}, fosA3, oqxAB, qnrS, qnrB</i>	<i>K. pneumoniae</i>	Fresh vegetables	China	Liu et al. (2019)
<i>mcr-1</i>	<i>bla_{NDM-5}</i>	<i>E. coli</i>	Wastewater treatment plant	China	Han,H et al. (2020)
<i>mcr-8</i>	<i>bla_{NDM}</i>	<i>K. pneumoniae</i>	Animals and humans	China	Wang X et al., 2018
<i>mcr-1</i>	<i>NDM-4</i>	<i>E. coli</i>	Hospitalized patients	Lebanon	Al-Bayssari et al. (2021)

et al., 2021). WHO guidelines recommend decreasing the use of antibiotics as feed additives and prophylactic agents in food producing animals that are critical for medical purposes (Aidara-Kane et al., 2018). Considering this, stringent measures have to be adopted to reduce the veterinary use of colistin since it is considered a last-resort antibiotic for the treatment of many infectious diseases in humans. High-income countries can assist LMICs in tackling colistin resistance by sponsoring financial assistance to improve biosecurity and biosafety in poultry settings and promote sustainable production of meat products (Mendelson et al., 2016). The present one health approach agenda of WHO have included guidelines for the use of antibiotics, focussing humans and food animals as important entities in combating AMR, but colistin resistance has become a serious issue for the environment, wildlife and aquaculture as discussed before. One health approach can be extended to these sectors as well to effectively tackle the issue of colistin resistance.

The success of South Africa's One Health stewardship programme can be considered as a model for other LMICs. As a part of the initiative, the veterinary use of colistin was reduced, AMR surveillance was strengthened and sludge management programs were implemented in farms (Mendelson et al., 2018). In order to prevent the abuse of colistin for medical purposes, prescriptions bearing the personal details of the physician, patient, institution along with the details of colistin product to be used and an informed consent of patient were mandated as part of this initiative in the country (Mendelson et al., 2018). This strategy can not only help in preventing the indiscriminate use of colistin but also, helps to monitor the development of AMR based on patient's response to colistin therapy. However, the initiative generated few data on environmental surveillance, which needs to be further developed since AMR from drinking water facilities in South Africa has been reported (Mendelson et al., 2018).

The High Council for Public Health (HCSP) in France have also taken a multistep initiative in 2016, to resolve colistin resistance by recommending four cardinal points. First suggestion focussing on the screening of all carbapenemase-producing Enterobacterales for *mcr-1* genes. The second recommendation focuses on hygiene precautions that needs to be observed. The third recommendation involves reporting *mcr-1/2* bearing strains to national reference center for antibiotic resistance. The final recommendation suggests the conduct of epidemiological surveys involving experts from HCSP, l'Observatoire National de l'Epidémiologie de la Résistance Bactérienne aux Antibiotiques,

national reference centre for antibiotic resistance (in Clermont-Ferrand) and "Santé publique" France for detecting the prevalence of *mcr-1* in Enterobacterales (Lepelletier et al., 2018). Action plans and initiatives taken at the national level are highly commendable, but colistin resistance is a global issue and requires international coordination, in the absence of which the response to the threat of AMR remains uneven and inadequate across countries (Laxminarayan et al., 2020).

The current COVID-19 pandemic scenario threatens the efforts taken in fighting AMR as it can contribute to the surge of AMR (Seethalakshmi et al., 2021). This arises due to the increase in the occurrence of secondary bacterial infections in COVID 19 patients coupled with the use of broad-spectrum antibiotics commonly among such patients (Clancy et al., 2020). Considering the scarcity of data, it can be expected that the use of antibiotics at present is rampant and undocumented (Miranda et al., 2020; Lucien et al., 2021). It is worth mentioning that colistin is also one of the antibiotics administered along with hydroxychloroquine in severe COVID-19 cases and colistin resistant bacteria was isolated from catheters of COVID-19 patient (Dimopoulos et al., 2020). Efforts to address the issue of colistin resistance should continue even amidst the pandemic. Global antimicrobial stewardship and antimicrobial surveillance policies should be upgraded and strengthened to cope with the present pandemic situation.

7. Future direction and conclusion

The alarming increase of colistin resistance worldwide intimates that we are on the brink of losing a last line antibiotic effective against the devastating carbapenem-resistant bacterial infections. Whilst the use of colistin as growth promoting agent is forbidden, their prophylactic still prevails and has to be monitored uniformly worldwide. For this development of regional databases tracking colistin usage can be valuable since it can make antimicrobial resistance surveillance easy and effective. Research indulged in studying the molecular mechanisms of colistin resistance needs to be encouraged as there are still gaps in our understanding of colistin resistance. While exploring alternative therapies will decrease our reliability on antibiotics and subsequent development of AMR (Luepke et al., 2017), discovering strategies to thwart the development of AMR can prevent the pipeline of antibiotic discovery from being futile. Discovery of antibiotics that have multiple targets rendering bacteriostatic or bactericidal activity could be another interesting approach to minimise the development of AMR. Repurposing of

non-antimicrobial drugs as antimicrobial agents is also a promising strategy to explore since bacteria have lesser tendency in developing resistance to these drugs due to their difference in targets compared to antibiotics (Seethalakshmi et al., 2020).

It will be a herculean task to control the selection and transmission of *mcr-1*, if the genes coexist with any other ARGs on the same plasmid (Yang et al., 2016). Hence effective management of household, hospital and industrial waste is the lone solution to restrain the circulation of *mcr-1* to different environments. Presence of *mcr-1* in patients without any history of colistin consumption, international travel, or contacts with animals is more worrying (Delgado-Blas et al., 2016) and epidemiological studies to trace the route of *mcr-1* transmission needs to be given more importance in the coming years. The link between AMR and continuous use of antibiotics remains unknown to a large sector of farmers even today and AMR is perceived as an inefficacy of the antibiotic in preventing infectious diseases rather than a result of anthropogenic activities amongst farmers (Albernaz-Gonçalves et al., 2021). To minimise the indiscriminate use of colistin in poultry and aquaculture farms, pharmacists should be instructed to provide the antibiotic only under the prescription of a veterinarian and awareness should be raised amongst farmers regarding the risks of AMR.

AMR surveillance programs focussed on water bodies needs to be established to monitor the flow of *mcr* to other ecosystems. The spread of *mcr* through underground water system also needs surveillance since it is an important reservoir of water for household and irrigation activities. Also, wells have short hydraulic retention time which might hasten the spread of resistant bacteria (Sun et al., 2017). Absence of a cheap and effective platform for detection of *mcr* and colistin residues further complicates the issue of colistin resistance because most of the detection methods that are currently employed may not be available to many LMIC countries. In the near future, a simpler and cost-effective method will be required for detection of colistin resistance and colistin residues, which is equally accessible to all countries across the globe.

CRedit authorship contribution statement

P.S. Seethalakshmi: Investigation, Visualization, Writing – original draft. **Riya Rajeev:** Writing – review & editing. **Anushara Prabakaran:** Investigation, Writing – review & editing. **George Seghal Kiran:** Supervision, Writing – review & editing. **Joseph Selvin:** Conceptualization, Supervision.

Data Availability

This is a review report, do not have research data.

Acknowledgements

This work has been supported by the Department of Biotechnology (DBT), Ministry of Science and Technology, Government of (DBT Reference: BT/IN/Indo-UK/AMR-Env/02/JS/2020-21).

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